

# Doubly Positive and Totally Positive Kernels

A compact account for positive sketching

**Terminology.** In the matrix literature, the standard names are *doubly nonnegative* and *completely positive*. Below I use *doubly positive* and *totally positive* for the analogous kernel notions, because the latter is what is needed for positive feature sketching. To avoid ambiguity, note that “totally positive” here does *not* mean Karlin’s total positivity, i.e. nonnegativity of all minors of a two-variable kernel; it means “completely positive type” in the sense of the completely positive matrix cone [Kar68, DGL21].

## 1 The finite-dimensional model

For a symmetric matrix  $A \in \mathbb{R}^{n \times n}$ , define

$$\text{DNN}_n := \{A \in \mathbb{R}^{n \times n} : A \succeq 0, A_{ij} \geq 0 \ \forall i, j\}$$

and

$$\text{CP}_n := \{A \in \mathbb{R}^{n \times n} : A = BB^\top \text{ for some } B \geq 0\}.$$

Equivalently,

$$\text{CP}_n = \text{cone}\{vv^\top : v \in \mathbb{R}_+^n\}.$$

Thus every completely positive matrix is doubly nonnegative:

$$\text{CP}_n \subseteq \text{DNN}_n.$$

The classical finite-dimensional fact is

$$\boxed{\text{CP}_n = \text{DNN}_n \text{ for } n \leq 4,} \quad \boxed{\text{CP}_n \subsetneq \text{DNN}_n \text{ for } n \geq 5.}$$

See, for example, the monograph of Berman and Shaked-Monderer [BSM03]; the  $5 \times 5$  difference is discussed in Anstreicher–Burer–Duer [ABD09].

## 2 Kernel analogues

Let  $X$  be a set and let

$$k : X \times X \rightarrow \mathbb{R}$$

be symmetric. For points  $x_1, \dots, x_n \in X$ , write

$$K[x_1, \dots, x_n] := (k(x_i, x_j))_{i,j=1}^n.$$

**Definition 1** (Doubly positive kernel). A kernel  $k$  is *doubly positive*, written  $k \in \text{DP}(X)$ , if

$$K[x_1, \dots, x_n] \in \text{DNN}_n$$

for every finite choice  $x_1, \dots, x_n \in X$ . Equivalently,  $k$  is positive semidefinite and pointwise nonnegative:

$$\sum_{i,j=1}^n c_i c_j k(x_i, x_j) \geq 0 \quad \text{for all } c_i \in \mathbb{R}, \quad k(x, y) \geq 0 \quad \forall x, y \in X.$$

**Definition 2** (Totally positive / completely positive-type kernel). A kernel  $k$  is *totally positive*, written  $k \in \text{TP}(X)$ , if

$$K[x_1, \dots, x_n] \in \text{CP}_n$$

for every finite choice  $x_1, \dots, x_n \in X$ .

The inclusion

$$\text{TP}(X) \subseteq \text{DP}(X)$$

is immediate from  $\text{CP}_n \subseteq \text{DNN}_n$ . Consequently, on a fixed sample of  $n \leq 4$  points the doubly positive and totally positive tests coincide, whereas from  $n = 5$  onward they can differ. The point is that  $\text{TP}(X)$  is the kernel-level condition corresponding to nonnegative Gram factorizations on every finite sample.

### 3 Why total positivity is the sketching condition

A positive sketch of  $k$  is an integral representation

$$k(x, y) = \int_Z \langle \phi(x, z), \phi(y, z) \rangle_{\mathbb{R}^p} d\mu(z), \quad \phi(x, z) \in \mathbb{R}_+^p. \quad (1)$$

This is the structure used by positive random features: the feature map takes values in the positive orthant, and finite random features approximate the integral by a finite sum [SC20].

Fix  $x_1, \dots, x_n \in X$ . For each coordinate  $a = 1, \dots, p$  and latent point  $z \in Z$ , set

$$v_{a,z} := (\phi_a(x_1, z), \dots, \phi_a(x_n, z))^\top \in \mathbb{R}_+^n.$$

Then (1) gives

$$K[x_1, \dots, x_n] = \sum_{a=1}^p \int_Z v_{a,z} v_{a,z}^\top d\mu(z).$$

Since  $\text{CP}_n$  is a closed convex cone, the matrix belongs to  $\text{CP}_n$ . Therefore

$$\boxed{\text{positive sketch} \implies k \in \text{TP}(X).}$$

Conversely, when  $X$  itself is finite, this condition is exactly sufficient:  $K \in \text{CP}_n$  means  $K = BB^\top$  with  $B \geq 0$ , so one can take the columns of  $B$  as positive features. For general spaces,  $\text{TP}(X)$  is the finite-restriction condition; analytic treatments often define the corresponding cone as the closure of conic combinations of rank-one kernels  $f \otimes f$  with  $f \geq 0$  [DGL21].

### 4 Complexity

For a fixed finite sample, checking  $K \in \text{DNN}_n$  is tractable: one checks entrywise nonnegativity and positive semidefiniteness. Checking  $K \in \text{CP}_n$  is very different. As  $n$  varies, the strong membership problem for the completely positive cone is NP-hard, and even weak membership for the completely positive cone and its dual copositive cone is NP-hard [DG14]. Thus, for finite data, membership in  $\text{TP}$  is exactly the computationally difficult part; for an infinite kernel, global membership in  $\text{TP}(X)$  asks for this condition on every finite restriction.

## 5 Examples

**Example 1** (Finite positive features). If  $h : X \rightarrow \mathbb{R}_+^r$  and

$$k(x, y) = \langle h(x), h(y) \rangle_{\mathbb{R}^r},$$

then  $k \in \text{TP}(X)$ , with the trivial sketch  $Z = \{*\}$  and  $\phi(x, *) = h(x)$ .

**Example 2** (Positive autocorrelations). Let  $X = \mathbb{R}^d$  and  $k(x, y) = \psi(x - y)$ . Suppose there are functions  $g_\ell \in L^2(\mathbb{R}^d)$ ,  $g_\ell \geq 0$ , such that

$$\psi(t) = \sum_{\ell} \int_{\mathbb{R}^d} g_\ell(u) g_\ell(u + t) du.$$

Then  $k \in \text{TP}(\mathbb{R}^d)$  because

$$k(x, y) = \sum_{\ell} \int_{\mathbb{R}^d} g_\ell(u - x) g_\ell(u - y) du,$$

so one may take  $z = (u, \ell)$  and

$$\phi(x, (u, \ell)) = g_\ell(u - x) e_\ell.$$

For instance,  $g = \mathbf{1}_{[0,1]}$  on  $\mathbb{R}$  gives the triangular kernel

$$k(x, y) = (1 - |x - y|)_+.$$

**Example 3** (Gaussian kernel). For

$$k_\varepsilon(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\varepsilon}\right),$$

set

$$g_\varepsilon(u) = \left(\frac{2}{\pi\varepsilon}\right)^{d/4} \exp\left(-\frac{\|u\|^2}{\varepsilon}\right).$$

Then

$$\int_{\mathbb{R}^d} g_\varepsilon(u - x) g_\varepsilon(u - y) du = \exp\left(-\frac{\|x - y\|^2}{2\varepsilon}\right),$$

and hence  $k_\varepsilon \in \text{TP}(\mathbb{R}^d)$ . This is the clean Lebesgue-measure version of the Gaussian positive-feature construction.

**Example 4** (Gaussian scale mixtures). If

$$k(x, y) = F(\|x - y\|^2), \quad F(s) = \int_0^\infty e^{-as} d\nu(a),$$

for a finite positive measure  $\nu$ , then  $k \in \text{TP}(\mathbb{R}^d)$  by mixing the Gaussian sketches. Equivalently, any completely monotone radial kernel  $F(\|x - y\|^2)$  is totally positive. This includes squared-exponential kernels, rational-quadratic kernels, inverse-multiquadrics, and Matern kernels via their standard Gaussian-mixture representations; see, e.g., Rasmussen–Williams for standard Gaussian-process covariance families [RW06].

## 6 Closure properties

**Proposition 1** (Sums and pointwise products). *Both  $\text{DP}(X)$  and  $\text{TP}(X)$  are stable under nonnegative sums and pointwise products. More precisely, if  $k_1, k_2 \in \text{TP}(X)$  and  $a_1, a_2 \geq 0$ , then*

$$a_1 k_1 + a_2 k_2 \in \text{TP}(X), \quad k_1 k_2 \in \text{TP}(X),$$

where  $(k_1 k_2)(x, y) = k_1(x, y)k_2(x, y)$ .

*Proof.* For sums, take the disjoint union of the latent spaces and scale the features by  $\sqrt{a_1}$  and  $\sqrt{a_2}$ . For products, if

$$k_i(x, y) = \int_{Z_i} \langle \phi_i(x, z_i), \phi_i(y, z_i) \rangle d\mu_i(z_i),$$

then

$$k_1(x, y)k_2(x, y) = \int_{Z_1 \times Z_2} \langle \phi_1(x, z_1) \otimes \phi_2(x, z_2), \phi_1(y, z_1) \otimes \phi_2(y, z_2) \rangle d\mu_1(z_1)d\mu_2(z_2),$$

and the tensor-product feature is still entrywise nonnegative. The same statements for  $\text{DP}(X)$  follow from the Schur product theorem and entrywise nonnegativity.  $\square$

## 7 Counterexamples

### A finite kernel: doubly positive but not totally positive

Let  $C_5$  be the cycle graph on five vertices and let

$$A_t = I_5 + t \text{Adj}(C_5).$$

The eigenvalues of  $\text{Adj}(C_5)$  are  $2 \cos(2\pi j/5)$ ,  $j = 0, \dots, 4$ ; hence  $A_t \succeq 0$  for  $0 \leq t \leq 1/\varphi$ , where  $\varphi = (1 + \sqrt{5})/2$ . Thus, for example,

$$A_{3/5} \in \text{DNN}_5.$$

But  $A_t \notin \text{CP}_5$  for  $t > 1/2$ . Indeed, if  $A_t = BB^\top$  with  $B \geq 0$ , then the zero entries of  $A_t$  force every column of  $B$  to be supported on a clique of  $C_5$ . Since  $C_5$  has no triangles, each column is supported on a single vertex or on one edge. To produce the off-diagonal value  $t$  on a fixed edge, the corresponding columns contribute at least  $2t$  to the two diagonal entries, by  $u^2 + v^2 \geq 2uv$ . Summing over the five edges gives

$$\text{tr}(A_t) \geq 10t.$$

But  $\text{tr}(A_t) = 5$ , so complete positivity would imply  $t \leq 1/2$ . Therefore  $A_{3/5}$  is doubly nonnegative but not completely positive. On the finite space  $X = \{1, \dots, 5\}$ , the kernel  $k(i, j) = (A_{3/5})_{ij}$  is doubly positive but not totally positive.

### A translation-invariant counterexample on $\mathbb{R}$

The gap is not merely a finite-space pathology. Define

$$\psi(t) = \frac{9}{10} \left[ \frac{\sqrt{5}}{5} + \frac{5 - \sqrt{5}}{5} \cos\left(\frac{2\pi t}{5}\right) + \frac{1}{9} \right], \quad k(x, y) = \psi(x - y).$$

This kernel is continuous, translation-invariant, positive definite, and strictly positive pointwise: it is a positive linear combination of  $1, e^{2\pi it/5}, e^{-2\pi it/5}$ , and its minimum is positive. Also  $\psi(0) = 1$ .

For  $x_j = j$ ,  $j = 0, \dots, 4$ , the Gram matrix is

$$K[x_0, \dots, x_4] = \frac{9}{10} \left( W + \frac{1}{9} J \right),$$

where  $J$  is the all-ones matrix and

$$W = \begin{pmatrix} 1 & a & 0 & 0 & a \\ a & 1 & a & 0 & 0 \\ 0 & a & 1 & a & 0 \\ 0 & 0 & a & 1 & a \\ a & 0 & 0 & a & 1 \end{pmatrix}, \quad a = \frac{\sqrt{5} - 1}{2}.$$

Let

$$H = \begin{pmatrix} 1 & -1 & 1 & 1 & -1 \\ -1 & 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 & 1 \end{pmatrix}$$

be the Horn copositive matrix. Every completely positive matrix  $B$  satisfies  $\langle H, B \rangle \geq 0$ , while

$$\left\langle H, \frac{9}{10} \left( W + \frac{1}{9} J \right) \right\rangle = \frac{9}{10} \left( 5(2 - \sqrt{5}) + \frac{5}{9} \right) < 0.$$

Hence this finite restriction is not completely positive, and  $k$  is doubly positive but not totally positive. Thus translation invariance and strict pointwise positivity do not by themselves imply positive sketchability.

## References

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